

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 1 of 15

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	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 2 of 15

INDEX

1.	OVERVIEW	3
2.	DISCUSSION	3
3.	DEFINITIONS	3
4.	WORKING PHASES	4
4.1	CODES FOR STATISTC ACTIVITIES	5
5.	MAIN OPERATIONS	6
5.1	WITH THE DRILLING RIG ON LOCATION.....	6
5.2	WITHOUT THE DRILLING RIG ON LOCATION	7
6.	DETAILS OF MAIN OPERATION	8
7.	TIME BREAKDOWN MATRIX	13

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 3 of 15

1. OVERVIEW

The Time breakdown procedure does not aim to translate into codes the operations performed on the well, but will be used as a tool to identify the “Main Operations” which were carried out with or without the rig.

These codes will be later used as a standard comparison tool for statistical analyses.

2. DISCUSSION

The Time Breakdown defines the different Working Phases of the well.

These working phases are grouped according to **Tab.- 4-1**.

In a second level, the different Working phases are divided into Main Operations which are specific for the two situations: the drilling rig is on location **Tab.- 5-1** and the drilling rig isn't on location **Tab.- 5-2**.

However, in order to limit the number of codes and to guide the user when he assigns the appropriate code while filling the daily operational reports, the Main Operations will be defined by their category (e.g.: Tripping), and not by the reason for their execution (e.g.: Trip for bit change).

From the combination of Main Operations and their Relevant Details, a “Time Breakdown Matrix” was created (Tab. - 7-1). This matrix will be used to identify the *activity codes* to be reported in the daily reports (ARPO 02).

Each combination is also characterized by an “Indicator” that determines if it belongs to “Planned Operations” (**P**), “Unplanned Operations” (**U**), “Trouble operation” (**T**) or to “Unplanned Trouble” (**X**)

3. DEFINITIONS

P = Planned: Time spent performing the operations which are a part of the original well plan or well objective and are included in the authorized budget (AFE).

Example: the time required to resolve the anticipated loss circulation problem or shut-off water flow.

U = Unplanned: All operations which are performed and were not included in the original well plan or well objective. Usually these operations are associated with a special request from the Exploration Department “Geology or Geophysics”.

Examples: run extra Wire Line logging, run additional casing string, changing the original well objective or deepening of a well beyond the original approved total depth.

T = Trouble: Any trouble that causes a time delay in the progress of a planned operation. It includes the time required to resolve the problem and reaching the same point or the depth at which the event occurred.

X = Unplanned Trouble: Refers to any occurrence that causes a time delay in the progress of an unplanned operation (as defined above). It includes the time required to resolve the problem and reaching the same point or the depth at which the event occurred.

Examples: pipe sticking during unplanned deepening operation, logging tools stuck or lost during unplanned logging operation.

N = Non-Productive Time

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 4 of 15

4. WORKING PHASES

PHASE	THE PHASE FORESEES THE OPERATION IN ORDER TO:	STARTING PHASE (date and time)	END PHASE (date and time)
RIG or Rigless Equipment MOVING	Move the rig or the equipment (in case of rigless operation) on the well location (Moving; Positioning; rig-up; etc).	Contract commencement.	End of rig or Rigless Equipment testing.
PRELIMINARY	Start Drilling, Completion and W.O. activities (Conductor pipe; Skidding; killing; etc).	End of rig testing.	Starting of the Re-Entry; Drilling; Workover phases.
RE-ENTRY	Restart the activity on an existing well (well head & B.O.P. operation, milling, plugging, etc.).	When the first tool is ready to be run in hole.	Run in hole of the first bit to start drilling.
DRILLING	Reach the planned depth	Run in hole of the first bit to start drilling. The start of each drilling phase is identified by the start of the run in hole of the first bit to drill the phase.	• Exploration wells: end recording of last log or reaching of final depth. • Development well: end of operation on the production casing with bit scraper on bottom.
WORKOVER	• Heavy: pull out the completion from the well and mechanically restore the well. • Light: Inspect, restore and modify the completion without pulling it out.	From the moment the first tool goes through the master valve.	• Heavy: end of restoration operation on the production casing with bit scraper on bottom. • Light: starting of well testing operation.
COMPLETION	Perform the configuration of well completion.	1. First completion: • Exploration wells: end recording of last log or reaching of final depth. • Development well: end of operation on the production casing with bit scraper on bottom. 2. Other completions: • End of restoration operation on the production casing with the bit scraper on bottom.	End make-up of production cross.

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 5 of 15

WELL TESTING	Perform well testing.	From starting run in hole well testing string.	• D.S.T.: up to starting of well abandoning operation, or starting of well completion phase, drilling restart. • Production test: from the well start-up or starting of well abandoning operation.
WELL ABANDONING	Carry out the permanent or temporary abandonment of the well.	From starting of run in hole D.P., wire line, coiled tubing for the positioning of the first plug (cement, bridge, etc.).	End of operation on the well
SECURE THE WELL	All the operation to secure the well (Run in hole plugs or B.p.v., redressing tsv or s.c.s.s.v., test s.c.s.s.v., etc)	From starting of run in hole the first tool	From the well start-up or starting of other working phases
PRODUCTION MAINTENANCE	All the operation to production maintenance (Gradient survey, acid job washing job, etc).	From starting of run in hole the first tool.	From the well start-up or starting of other working phases.

Tab.- 4-1

4.1 CODES FOR STATISTC ACTIVITIES

The codes in the following **Tab.- 4-2** must not be used during the well operation but only for Statistic Activity.

DOWN TIME (Details from 21 to 26)	Stops not due to incidents including well safety operation (if carried out).	From interruption of operation underway on the well.	Up to restarting of planned operation.
INCIDENT (Operation flagged with T or X)	Solve the well incident that stops the planned or Unplanned operation, including all operation required to restore the conditions existing before the incident.	From the time the event occurs.	Up to the moment well conditions existing before the event are restored.

Tab.- 4-2

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 6 of 15

5. MAIN OPERATIONS

5.1 WITH THE DRILLING RIG ON LOCATION

The following main operations can be used also in case of rigless operations.

MAIN OPERATIONS	DESCRIPTION
MOVING Rig up/Rig down	All the time spent in moving the rig or the rig-less equipment from the base to location or from a location to another location; it also includes: the time for all the operations for rigging-up/rigging down, the time for the preparation of the rig site (positioning, anchoring, jacking up/down, deballasting) and the time spent in testing the rig or the equipment before the starting of the assigned activity.
SKIDDING	Skidding Rig operations on a new well or on a new cellar. In case of skidding between two wells of a same cluster, preparation time must be charged to the well under way, while skidding times shall be charged to the arrival well.
WELL HEAD	All the time spent to perform any operations on the well head (i.e. welding, installing new spools & seals); it also includes all the time necessary for the preparation of the wellhead, assembly, recovery of well head and all relevant tests.
B.O.P.	All the time spent to perform works in the BOP stack such as: P/U new BOP stack, L/D BOP stack, handling of B.O.P., changing pipe rams and all the relevant tests.
CONDUCTOR PIPE	All the time spent in driving or drilling to set the conductor pipe; it includes the time required to clean the CP to allow for more penetration.
DRILLING	All the operations involved in drilling the hole according to the approved well plan; it also includes: rotation time, steering, circulation, reaming, etc..
SIDE TRACK or RE-DRILLING	All the time spent to sidetrack the well (due to unforeseen accident in the well) until reaching the same TVD depth before the accident. In some specific cases, when the sidetrack is executed after reaching the final well target, the TVD of the sidetrack can be lower than the previous one. The phase will start when the bit is run in the hole to dress the cement plug, also the phase includes the integrity test.
CASING JOB	All the time spent to: run & set a casing string, liner, tie back & tubing. The phase will end upon starting to pump the cement spacer fluid.
CEMENTING JOB	All the time spent in: cementing or re-cementing (remedial cementing job) of casing, liner, or tie-back; it also includes the time for plugging and squeezing cementing in well. The phase will last from the time of pumping the cement spacers to the end of W.O.C. period.
CORING	All the time spent to cut & recover a core, including the time spent in making-up and running in hole the core barrel, and cutting the core. The phase also includes the intermediate wiper trips/reaming operations between a core and another one (in case of continuous coring) .
LOGGING	All the time spent in logging operations, including the time spent to R/U & R/D Electrical logging equipment, recording time, tripping time before and after logging operations, intermediate trips required to condition the hole and mud to resume the logging operations - the fishing of logging tools.

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 7 of 15

WELL COMPLETION	All the operations required to run a completion string: the phase will last from the end of the operations conducted in the production casing “with bit/scrapper on bottom” , to the starting of the well testing or rig down.
WELL DECOMPLETION	All the operations necessary to perform the pull out a completion string: from end of B.O.P. installation, to the end of extraction of the last completion device pulled out.
WELL PREPARATION	All the operations necessary for the mechanical restoration of the well after pulling out the completion: from the end of completion removal, to the end of restoration operations on the production casing with bit or scrapper on bottom.
SAND CONTROL	All the operations necessary to perform well intervention able to control sand coming from the formation (gravel pack, etc.): from preparation to well intervention, to the test for the check of the gravel top.
WELLBORE TREATMENT	All the operations necessary to perform stimulation, acidification, fracturing, etc.: from rig up of the pumping unit, to starting of preparation for well testing.
WELL TEST &/or D.S.T.	All the operations necessary to enable the well to flow: from the rig up of equipment for well test, to rig down.
WELL ABANDONING	All the operations involved in performing the permanent or temporary abandoning of the well. The phase starts from the time of running in hole (with the DP or tubing stinger, wire line & coiled tubing) to set the first plug (cement, bridge plug, etc.), and ends after securing the well before leaving the location.

Tab.- 5-1

5.2 WITHOUT THE DRILLING RIG ON LOCATION

The following main operations, are defined to coriedout specific performance analysies for the operations with slike line, coiled tubing and in case of artificial lift.

They shall be used only in case of rigless operations.

SLIKE LINE	All the operation performed with slike line equipment.
COILED TUBING	All the operation performed with coiled tubing equipment.
ARTIFICIAL LIFT	All the operations necessary to install or maintenance all the artificial lift equipment.

Tab.- 5-2

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 8 of 15

6. DETAILS OF MAIN OPERATION

General Definition

P = Planned Operations;

U = Unplanned Operations.

T = Trouble

X = Unplanned trouble

(as defined above)

[1] Preparation: Time actually spent in the preparation operations, as described in the Main Operation section.

(e.g.: A1-P = preparation of rig move;
B1-P = preparation of skidding the rig;
G1-P = preparation (M/U Tubing string in stands) for well testing operation ;
O1-P = preparation of well bore treatment; etc.).

[2] Rig or Equipment Move: Time actually spent in moving the rig (Moving & Skidding).

(e.g.: A2-P = moving;
B2-P = skidding).

[3] Rotation Hours on Bottom: Hours actually spent in drilling (rotation hours of the bit & coring); it also includes the rotations hours to drill the surface section for the conductor pipe, pilot hole, deepening hole for gravel pack, sidetracking and coring.

(e.g.: E3-P = Drilling;
I3-P = Coring;
D3-P= Drilling surface hole to set C.P.
E3-U = Deepening the well after reaching the planned Total Depth as per request of the exploration department.
F3-T = Side-track the well after stuck pipe incident;
F3-X = Side-track the well after stuck pipe incident in an unplanned deepening operation etc.)

[4] Trips: Hours actually spent to run in hole/pull out of hole any object from the well (B.H.A, Casing, Conductor Pipe, Completion string, Logs, Wire line, Coiled tubing, etc.).

(e.g.: G4-P = casing running;
I4-P = core barrel running;
K4-P = completion running;
L4-U = completion pull out for equipment replacement).

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 9 of 15

[5] Installation/ Disassembly: Hours actually spent for rig-up/rig down or equipment make-up or lay down.

(e.g.: B5-P = rig-up;

J5-P= rig up logging unit;

J5-T= rig up logging unit to execute back-off ;

O5-P = rig up snubbing unit).

[6] Wiper Trip and Reaming: Hours actually spent in tripping (wiper or short trip) , including reaming time to reach the hole bottom.

(e.g.: T-E6 = reaming;

P-E6 = wiper trip before tubing).

[7] Circulation: Hours actually spent in circulation (intermediate, on bottom, to control a kick, bottoms up, etc.).

(e.g.: E7-P = circulation bottoms up for sampling;

E7-T = kick control during drilling operation:

I7-T = kick control during logging operation:

G7-P = intermediate and on bottom circulation during tubing).

[8] Fluids Preparation: Hours actually spent in fluids preparation (mud, pills, slurry, etc.).

(e.g.: E8-P = mud mixing;

E8-T = preparation of intervention pills;

N8-P = slurry mixing for gravel pack;

H8-P = cement/spacer preparation during cementing job).

[9] Fluids Pumping in Well: Hours actually spent for pumping fluids in well.

(e.g.: E9-T = pills pumping;

H9-P = pumping of slurry and spacers).

[10] Fluids/Core Recovery (cores, D.S.T., soil test): Hours actually spent for the recovery of the core, fluids from D.S.T., etc..

(e.g.: I10-P = core recovery;

P10-P = fluid recovery from D.S.T.;

B10-P = cores for soil test).

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 10 of 15

[11] Measurement in Well: Hours actually spent in performing or recoding some measurement in the well such as: Electrical logs, survey hole, flow check, L.O.T., F.I.T, recording build-up pressure during well testing , etc..

(e.g.: J1-P = logging;
J11-U = logging in an unplanned deepening operation;
E11-P = survey;
P11-P = build-up pressure during well testing).

[12] Equipment Test: Hours actually spent in testing equipment such as: B.O.P., well head, down hole motors, tools in general, etc..

(e.g.: C12-P = B.O.P. test;
E12-P = test of down hole motors;
H12-P = line test;
K12-P = packer test).

[13] Working-on/Fishing: Hours actually spent to work on some tools in well, such as: Jar & Bumper, Equipment Malfunctioning (D.V. collar, liner hanger, packer, bottom valves, etc.), Tool Positioning and Removal (plug, packer, nipple, valve, etc.), etc., trip time included.

(e.g.: H13-T = attempt to open D.V. collar;
G13-T = attempt to set liner hanger;
K13-T = attempt to open SSD valve;
L13-T = attempt to pull-out SSR).

[14] Milling: Hours actually spent in milling or drilling out equipment's, such as: float collar, shoe, dressing top liner, milling material in well, casing mill-out, tools in general, etc..

(e.g.: G14-P = drill-out float collar and shoe;
G14-U = unplanned casing mill-out for side track to reach new target;
G14-T = casing mill-out for side-track due to stuck-pipe;
M14-P = scraping in the production casing).

[15] Through Tubing Operations: Operations made on completed well with coiled tubing, slick line, electric line, etc..

(e.g.: K15-P = string calibration;
O15-P = lifting operations).

[16] Shoot/Perforating: Hours actually spent for perforating/shooting, including trip time.

(e.g.: E16-T = clear bit nozzles;
E16-T = drill string back-off;
P16-P = casing and tubing perforation).

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 11 of 15

[17] Plugs and Squeezes: Hours actually spent, trip time included, to set plugs and squeeze any kind of fluids.

(e.g.: Q17-P = plugs for permanent abandoning;
H17-T = re-cementing with squeeze).

[18] Casing & Tubing Cut and Recovery: Hours actually spent to cut & recover casing string in hole

(e.g.: F18-U = unplanned casing cut and recovery for side track the well to resume the drilling operation;
F18-T = casing cut and recovery to side track the well due to well problem;
Q18-P = casing cut and recovery for well abandoning).

[19] Various Operations in Well: Hours actually spent in performing operations in the well not otherwise specified.

[20] Various Operations at Surface: Hours actually spent performing operations at surface not otherwise specified.

[21] Waiting in general: Hours actually spent for general waiting (W.O.C., pill effect, treatments, etc.), including the hours spent to put the well in safety conditions and resume operations.

(e.g.: G21-P = W.O.C.;
E21-T = waiting pill effect).

[22] Waiting on Environmental Conditions: Hours actually spent in waiting on environmental conditions (W.O.W., daylight, etc.), including the hours spent to put the well in safety conditions and resume operation.

(e.g.: E22-T = waiting on weather conditions in drilling;
K22-T = waiting on weather conditions in completion;
P22-P = waiting on daylight to open a well during the well testing.

[23] Waiting on Materials, Services, Personnel (Company's): Waiting on Materials, Services, Personnel of the Company, including the hours spent to put the well in safety conditions to resume the operations.

[24] Waiting on Materials, Services, Personnel (Contractor's): Waiting on Materials, Services, Personnel of the Contractor/s, including the hours spent to put the well in safety conditions to resume the operations.

	TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS	
Rev.0	Document No:	Page 12 of 15

[25] Maintenance and Repair: Including possible hours spent to put the well in safety conditions to resume the operations.

[26] Strike: Including possible hours spent to put the well in safety conditions to resume the operations.

[27] Not accurate or no news available: Used only in non-operated well

[28] Hole Opening : Hours actually spent in hole enlargement (rotation hours off bottom)

[29] Change Production Level: Hours actually spent performing changes production level or partialization.

(e.g.: R30-P = Change Production Level with slike line equipment;
S30-P = Change Production Level with coiled tubing equipment).

[30] Redress T.S.V.: Hours actually spent to pull out of hole, redress and run in hole tubing safety valves.

(e.g.: R31-T = Redress T.S.V. with slike line equipment;
S31-T = Redress T.S.V with coiled tubing equipment).

[31] Redress S.C.S.S.V.: Hours actually spent to pull out of hole, redress and run in hole surface controlled subsurface safety valves.

(e.g.: R32-T = Redress S.C.S.S.V with slike line equipment;
S32-T = Redress S.C.S.S.V with coiled tubing equipment).

[32] Test S.C.S.S.V. Hours actually spent to record the test of the surface controlled subsurface safety valves.

[33] Well Control Hours actually spent to record the test of the surface controlled subsurface safety valves.



**TIME DISTRIBUTION PROCEDURE
FOR DRILLING & COMPLETION OPERATIONS**

Rev.0

Document No:

Page 13 of 15



TIME DISTRIBUTION PROCEDURE FOR DRILLING & COMPLETION OPERATIONS

Rev.0

Document No:

Page 14 of
15

7. Time Breakdown Matrix

ACTIVITY	CODE	Preparation																																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
MOVING (Rig up/Rig down)	A	A1	A2						A4		A11	A12				A15	A16			A20	A21	A22	A23	A24	A25	A26	A27	A28						A32
SKIDDING	B	B1	B2			B5				B10		B12								B20	B21	B22	B23	B24	B25	B26	B27	B28						
WELL HEAD & B.O.P.	C	C1	C2		C4	C5		C7	C8	C9		C12	C13						C17		C20	C21	C22	C23	C24	C25	C26	C27	C28					C32
CONDUCTOR PIPE	D	D1			D4	D5		D7			D11	D12	D13	D14						D18	D20	D21	D22	D23	D24	D25	D26	D27	D28					
DRILLING	E	E1			E3	E4	E5	E6	E7	E8	E9	E11	E12	E13	E14		E16	E17		E19	E20	E21	E22	E23	E24	E25	E26	E27	E28					
SIDE TRACK OR RE-DRILLING	F	F1			F3	F4	F5	F6	F7	F8	F9	F11	F12	F13	F14		F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26	F27	F28					
CASING JOB	G	G1			G4	G5	G6	G7	G8	G9		G12	G13	G14				G17		G19	G20	G21	G22	G23	G24	G25	G26	G27	G28					
CEMENTING JOB	H	H1	H2		H4	H5	H6	H7	H8	H9		H12	H13				H16	H17		H19	H20	H21	H22	H23	H24	H25	H26	H27	H28	H29				
CORING	I	I1			I4	I5	I6	I7	I8	I9	I10	I12	I13	I14			I16	I17		I19	I20	I21	I22	I23	I24	I25	I26	I27	I28					
LOGGING	J	J1	J2		J4	J5	J6	J7	J8	J9		J11	J12	J13		J15	J16			J19	J20	J21	J22	J23	J24	J25	J26	J27	J28					
WELL COMPLETION	K	K1	K2		K4	K5		K7	K8	K9		K11	K12	K13		K15	K16	K17	K18	K19	K20	K21	K22	K23	K24	K25	K26	K27	K28					
WELL DECOMPLETION	L	L1	L2		L4	L5		L7	L8	L9		L11	L12	L13	L14	L15	L16	L17	L18	L19	L20	L21	L22	L23	L24	L25	L26	L27	L28					
WELL PREPARATION	M	M1			M4	M5	M6	M7	M8	M9		M11	M12	M13	M14		M16	M17	M18	M19	M20	M21	M22	M23	M24	M25	M26	M27	M28					
SAND CONTROL	N	N1	N2		N4	N5	N6	N7	N8	N9		N11	N12	N13		N15	N16	N17		N19	N20	N21	N22	N23	N24	N25	N26	N27	N28					
WELLBORE TREATMENT	O	O1	O2		O4	O5		O7	O8	O9		O11	O12	O13		O15	O17			O19	O20	O21	O22	O23	O24	O25	O26	O27	O28					
WELL TEST & D.S.T.	P	P1	P2		P4	P5		P7	P8	P9	P10	P11	P12	P13		P15	P16	P17		P19	P20	P21	P22	P23	P24	P25	P26	P27	P28	P29				
WELL ABANDONING	Q	Q1	Q2		Q4	Q5	Q6	Q7	Q8	Q9		Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28					
SLUKE LINE	R	R1	R2					R7	R8	R9		R11	R12				R15	R16	R17		R20	R21	R22	R23	R24		R26					R29	R30	R31
COILED TUBING	S		S2					S7	S8	S9		S11	S12			S14	S15	S16	S17	S18	S20	S21	S22	S23	S24		S26					S29	S30	S31
ARTIFICIAL LIFT	T	T1	T2					T7	T8	T9		T11	T12			T14	T15	T16	T17	T18	T20	T21	T22	T23	T24		T26					T29	T30	T31

Codes given in the matrix of Tab. 3 are not firmly fixed, therefore any other combination is possible

The last three activity shall be used only for rig less operations

	Geological Forecast for Development Well SPD21-12, Phase 21	 N.I.O.C. <i>Nars Oil & Gas Company</i>
Rev.01	Document No. SP2021- DR – 112– 860- GF – 0001	Page 15 of 15

Tab.-7-1